

Solve the 8 Queens Problem

PROGRAM:

N = 8

```
def print_solution(board):
    for row in board:
        print(" ".join("Q" if col else "." for col in row))
    print("\n")

def is_safe(board, row, col):
    for i in range(row):
        if (board[i][col] or
            (col - (row - i) >= 0 and board[i][col - (row - i)]) or
            (col + (row - i) < N and board[i][col + (row - i)])):
            return False
    return True

def solve_n_queens(board, row):
    if row == N:
        print_solution(board)
        return True

    found = False

    for col in range(N):
        if is_safe(board, row, col):
            board[row][col] = 1
            found = solve_n_queens(board, row + 1) or found
            board[row][col] = 0

    return found

board = [[0] * N for _ in range(N)]

solve_n_queens(board, 0)
```

OUTPUT:

Q
 Q
 Q
 Q
 Q
 Q .
 . Q
 Q .
 . Q
 Q

Q
 Q
 Q
 Q
 Q .
 Q
 . Q
 Q

Q
 Q .
 Q
 Q
 Q
 . Q
 Q
 Q

